

Machine Learning Assignment- 36

1) Differentiate between :

- `model.fit()`
- `model.predict()`

Explain which one is used during training and which during testing.

Ans : • `model.fit()` -

This is used to train the Machine Learning model. During this step, the model learns the relationship between the input features (`X_train`) and the target labels (`Y_train`). It analyzes the training data, finds patterns, and stores the learned information inside the model. This method is called only once or whenever we want to retrain the model.

• `model.predict()` -

It is used to make predictions using the trained model. It takes new input data (usually `X_test` or any unseen data) and predicts the output based on what it learned during training. It does not learn anything new; it simply uses the knowledge already stored in the model.

• During training, we use:

`model.fit(X_train, Y_train)`

because the model is learning from the training data.

• During testing, we use:

`model.predict(X_test)`

because the model predicts the output for the test data.

2) Define Accuracy.

Write the mathematical formula for accuracy and explain it in simple words.

Ans: Accuracy is a performance metric that tells us how many predictions made by the model are correct out of the total predictions. It is usually expressed as a percentage.

Formula -

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100$$

$$= \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

where, TP is True Positive

TN is True Negative

FP is False Positive

FN is False Negative

Simple Explanation -

Suppose a model predicts the results for 100 samples.

- If 90 predictions are correct, then

$$\text{Accuracy} = \frac{90}{100} \times 100 = 90\%$$

Therefore

This means the model correctly predicted 90 out of 100 samples.

- 3) If a model predicts 80 correct results out of 100 samples:
- What is the accuracy?
 - Is high accuracy always good? Explain with reasoning.

Ans:

a) Using the formula,

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

$$= \frac{80}{100} \times 100 = 80\%$$

Therefore, the model's accuracy is 80%.

b) No, high accuracy is not always good.

A model may achieve high accuracy but still perform poorly in real-world situations, especially when the dataset is imbalanced.

Example -

Suppose a dataset contains 100 emails:

- 95 are Not Spam
- 5 are Spam

If a model predicts every email as "Not Spam", then it correctly predicts 95 emails.

$$\text{Accuracy} = 95\%$$

Although the accuracy is very high, the model fails to identify all spam emails, making it unreliable for spam detection.

Therefore, accuracy alone is not enough to judge a model. We should also consider other evaluation metrics such as Precision, Recall, and F1 & F1-Score, especially for imbalanced datasets.

4) Explain the concept of a Confusion Matrix. Why is it more informative than accuracy?

Ans : A Confusion Matrix is a table used to evaluate the performance of a classification model. It compares the actual values with the predicted values and shows how many predictions were correct and how many were incorrect.

It consists of four values :

- True Positive (TP)
- True Negative (TN)
- ~~True~~ False Positive (FP)
- False Negative (FN)

A confusion matrix is more informative than accuracy because accuracy only tells us the percentage of correct predictions. It does not tell us what type of mistakes the model made.

The confusion matrix shows :

- How many positive cases were predicted correctly.
- How many negative cases were predicted correctly.
- How many false alarms the model made.
- How many actual positive cases the model missed.

This helps ^{us} ~~up~~ understand the strengths and weakness of the model much better than accuracy alone.

5) In the Ball classification case study (Cricket Vs Tennis):
Explain clearly:

- True Positive
- True Negative
- False Positive
- False Negative

Give real meaning using the ball example.

Ans: Lets assume:

- Positive Class = Cricket Ball
- Negative Class = Tennis Ball

a) True Positive (TP)

A True Positive means the ball is actually a Cricket Ball, and the model also predicts it as a Cricket Ball.

Example -

- Actual Ball = Cricket Ball
- Predicted Ball = Cricket Ball The prediction is correct

Means Actual ^{Cricket} Ball $\rightarrow$ Predicted Cricket Ball = True Positive

b) True Negative (TN)

A True Negative means the ball is actually a Tennis Ball, and the model predicts it as a Tennis Ball.

Example -

- Actual Ball = Tennis Ball
- Predicted Ball = Tennis Ball

The prediction is correct

Means, Actual Tennis Ball $\rightarrow$ Predicted Tennis Ball = True Negative

c) False Positive (FP)

A False Positive means the ball is actually a Tennis Ball, but the model incorrectly predicts it as a Cricket Ball.

This is called a False Alarm.

Example -

- Actual Ball = Tennis Ball
- Predicted Ball = Cricket Ball

The Prediction is wrong.

Means, Actual Tennis Ball $\rightarrow$ Predicted Cricket Ball = False Positive

d) False Negative (FN)

A False Negative means the ball is actually a Cricket Ball, but the model incorrectly predicts it as a Tennis Ball.

This means the model missed the Cricket Ball.

Example -

- Actual Ball = Cricket Ball
- Predicted Ball = Tennis Ball

The prediction is wrong.

Means, Actual Cricket Ball $\rightarrow$ Predicted Tennis Ball = False Negative.

6) If a tennis ball is predicted as Cricket ball:

- What type of error is this?

Ans: This is called a False Positive (FP).

- Why is it called that?

Ans: Let's Assume!

- Positive = Cricket Ball
- Negative = Tennis Ball

In this case:

- The actual ball is a Tennis Ball (Negative)
- But the model predicts it as a Cricket Ball (Positive).

Since the prediction is Positive, but the actual answer is Negative, it is called a False Positive.

It is also known as a False Alarm because the model incorrectly identifies a negative object as positive.

Example -

- Actual Ball : Tennis Ball
- Predicted Ball : Cricket Ball

This is a False Positive (FP).

7) In a confusion matrix:

	Predicted Cricket	Predicted Tennis
Actual Cricket	40	5
Actual Tennis	3	52

Calculate:

- Accuracy
- Total Errors
- Total Correct Predictions

Ans: a) Accuracy -

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

$$\text{Total predictions} = 40 + 5 + 3 + 52 = 100$$

$$\text{Accuracy} = \frac{92}{100} \times 100 = 92\%$$

Answer : 92%.

b) Total Errors -

$$\begin{aligned}\text{Errors} &= \text{FP} + \text{FN} \\ &= 3 + 5 = 8\end{aligned}$$

Answer = 8

c) Total Correct Predictions -

$$\begin{aligned}\text{Correct Predictions} &= \text{TP} + \text{TN} \\ &= 40 + 52 = 92\end{aligned}$$

Answer = 92

8) What is the difference between:

- Binary Classification
- Multiclass Classification

Give one real-world example for each.

Ans: Binary Classification is a type of classification where the model predicts only two possible classes. The output is either one class or the other.

Example:

Email Spam Detection

- Spam
- Not Spam

Multiclass Classification is a type of classification where the model predicts more than two classes.

Example:

Flower Classification (Iris Dataset)

- Setosa
- Versicolor
- Virginica

9) Explain how confusion matrix changes in multiclass classification compared to binary classification.

Ans: In binary classification, there are two classes, so the confusion matrix is a 2×2 table. It contains four values:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

In multiclass classification, there are more than two classes, so the confusion matrix becomes an $N \times N$ table, where N is the Number of classes.

For example:

- 3 classes $\rightarrow 3 \times 3$ confusion matrix
- 4 classes $\rightarrow 4 \times 4$ confusion matrix

Each row represents the actual class, and each column represents the predicted class.

The diagonal elements represent correct predictions, while the off-diagonal elements represent incorrect predictions.

10) In the Iris dataset:

- What are the features?

Ans: The iris dataset has four features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

These are the input variables (x) used to identify the flower species.